



UIL Computer Science Competition

State 2021

JUDGES PACKET - CONFIDENTIAL

I. Instructions

1. The attached printouts of the judge test data are provided for the reference of the contest director and programming judges. Additional copies may be made if needed for this purpose.
2. This packet must remain CONFIDENTIAL. Additional copies may be made and returned to schools when other confidential contest material is returned.

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Problem #1
60 Points

1. Adriana

Program Name: Adriana.java Input File: None

Test Output To Screen:

```
0000
0001
0011
0010
0110
0111
0101
0100
1100
1101
1111
1110
1010
1011
1001
1000
```

Problem #2
60 Points

2. Bruno

Program Name: Bruno.java

Input File: bruno.dat

Test Input File:

Lesser snow geese breed along Queen Maud Gulf, Baffin Island, Banks Island and Victoria Island, in the Northwest Territories and on Hudson Bay. They nest on low, grassy tundra plains and broad, shallow rivers near the coast, and on islands within shallow inland lakes. Lesser snow geese nest in colonies and lay an average of 4 to 5 eggs. Lesser snow geese have two color phases, a dark, blue, plumage and a white, snow, plumage. The two color phases are variations within the same race and do not indicate separate races. The sexes are similar in appearance in both phases, but the female is often smaller. Lesser snow geese can hybridize with Ross geese, which are similar in appearance. They have pinkish bills with black grinning patches, and the feet and legs are reddish pink. In the dark phase they have white heads and upper necks, with bluish gray bodies. In the white phase they are completely white except for black wing tips. The head can be stained rusty brown from minerals in the soil where they feed. They are very vocal and can often be heard from more than a mile away.

Test Output To Screen:

193

Problem #3
60 Points

3. Delfina

Program Name: Delfina.java

Input File: delfina.dat

Test Input File:

12	5 1	4
6	5 3	0 0
0 1	3 1	5 5
1 3	7	2 2
3 4	5 5	7 7
4 2	7 3	5
3 1	9 5	5 5
1 0	14 5	5 10
4	14 0	8 8
0 0	0 0	2 8
3 4	0 5	0 0
8 4	7	5
5 0	5 5	5 5
4	3 7	6 11
3 3	5 9	8 8
5 3	5 14	2 8
5 1	0 14	0 0
3 1	0 0	5
4	5 0	5 5
3 3	5	5 10
5 3	0 0	8 5
3 1	5 10	2 8
5 1	10 0	0 0
4	0 5	
3 3	10 5	

Test Output To Screen:

Simple
Simple
Simple
Self intersecting
Self intersecting
Simple
Simple
Self intersecting
Self intersecting
Self intersecting
Self intersecting
Self intersecting

Problem #4
60 Points

4. Eszter

Program Name: Eszter.java

Input File: eszter.dat

Test Input File:

```
10
{1.00}
{1.00,2.00}
{.99,2.00}
{1.00,1.99}
{1.87,8.34,12.18,12.30,20.54,21.98,38.93,44.41,53.90,55.08,57.81,59.80,61.38,69.43,72.78,
 74.04,74.55,75.41,96.67,100.00}
{5.13,20.03,24.81,27.05,48.76,55.79,59.52,61.35,61.55,62.69,66.22,71.24,74.02,82.22,87.14
,89.25,91.91,94.65,99.62,99.72}
{1.00,2.00,3.00,4.00,5.00,6.00,7.00,8.00,9.00,10.00}
{10.00,9.00,8.00,7.00,6.00,5.00,4.00,3.00,2.00,1.00}
{1.00,1.00,1.00,1.00,1.00,1.00,1.00,1.00,1.00,1.00,1.00,1.00,1.00,1.00,1.00,1.00}
{3.00,5.00,8.00,9.00,10.00,17.00,17.00,20.00}
```

Test Output To Screen:

```
Maximum profit is $1.00
Maximum profit is $2.00
Maximum profit is $2.00
Maximum profit is $2.00
Maximum profit is $116.14
Maximum profit is $200.30
Maximum profit is $10.00
Maximum profit is $100.00
Maximum profit is $16.00
Maximum profit is $24.00
```

Problem #5
60 Points

5. Hachiro

Program Name: Hachiro.java

Input File: hachiro.dat

Test Input File:

```
30
Lions 3 Tigers 2
Tigers 4 Eagles 1
Bulldogs 1 Tigers 1
Tigers 2 Panthers 2
Wildcats 0 Tigers 4
Lions 5 Eagles 1
Bulldogs 0 Lions 0
Lions 1 Panthers 0
Wildcats 2 Lions 1
Eagles 1 Bulldogs 3
Panthers 3 Eagles 2
Eagles 4 Wildcats 4
Panthers 1 Bulldogs 1
Bulldogs 5 Wildcats 4
Wildcats 2 Panthers 1
```

```
Tigers 6 Lions 1
Eagles 3 Tigers 4
Tigers 2 Bulldogs 3
Panthers 0 Tigers 0
Tigers 5 Wildcats 4
Lions 1 Eagles 0
Bulldogs 0 Lions 3
Panthers 5 Lions 1
Lions 3 Wildcats 3
Bulldogs 3 Eagles 2
Eagles 6 Panthers 5
Wildcats 2 Eagles 1
Bulldogs 4 Panthers 5
Wildcats 3 Bulldogs 5
Panthers 3 Wildcats 3
```

Test Output To Screen:

```
6A District 33 Soccer Table
Team      W  T  L  P  GD
Tigers    5  3  2 18  12
Bulldogs  5  3  2 18   3
Lions     5  2  3 17   0
Panthers  3  4  3 13   3
Wildcats  3  3  4 12  -5
Eagles    1  1  8  4 -13
```

Problem #6
60 Points

6. Irene

Program Name: Irene.java

Input File: irene.dat

Test Input File:

10	.	
3 9	.	
.....	.	
.....	.	
.....	10 2	
3 9	.*	10 10
.*...*.*.	*.	
..*...**.	.*	
*..**.*.*	*.	
3 3	.*	
...	*.	
...	.*	
...	*.	
6 4	..	
....	*.	
..*.	6 6	
*...		10 10
....		*.....
....		...*.....
....		*
10 1		*
.		*.....*
.	9 9	*
.		
.		...*
.		
.		**.*

Test Output To Screen:

```

1735
0
9
404
27
1
145871
5165306431734
9876298714905341
1300302342293
    
```

Problem #7
60 Points

7. Joseph

Program Name: Joseph.java

Input File: joseph.dat

Test Input File:

```
TRQ CTU QOI JYN PEV NIL BZZ VAS EBC ROK ABC BCD CDZ XYZ
51 1 50 100 75 12 62 43 38 8 27 81 95 59
ALL STR DJP FBO YWA RIK OOO WFS ZZT LOK KAR
38 19 60 70 79 9 51 88 60 50 51
```

Test Output To Screen:

```
EBC ALL 0
QOI LOK 0
TRQ KAR 0
TRQ OOO 0
QOI KAR 1
QOI OOO 1
ROK RIK 1
TRQ LOK 1
XYZ DJP 1
XYZ ZZT 1
BCD YWA 2
BZZ DJP 2
BZZ ZZT 2
NIL RIK 3
PEV YWA 4
PEV FBO 5
VAS ALL 5
BCD WFS 7
CDZ WFS 7
NIL STR 7
VAS LOK 7
ABC STR 8
BZZ FBO 8
CTU RIK 8
VAS KAR 8
VAS OOO 8
XYZ KAR 8
XYZ OOO 8
TRQ DJP 9
TRQ ZZT 9
XYZ LOK 9
QOI DJP 10
QOI ZZT 10
```


Problem #8
60 Points

8. Khushi

Program Name: Khushi.java

Input File: khushi.dat

Test Input File:

```
12
12.18 20.00
10.00 20.00
5.50 5.50
0.53 1.00
3.59 10.00
13.59 50.00
0.26 1.00
41.00 100.00
99.95 100.00
0.59 1.00
19.25 20.00
19.01 20.00
```

Test Output To Screen:

```
1-$5 2-$1 3-Quarters 1-Nickel 2-Pennies
1-$10
No Change Owed
1-Quarter 2-Dimes 2-Pennies
1-$5 1-$1 1-Quarter 1-Dime 1-Nickel 1-Penny
1-$20 1-$10 1-$5 1-$1 1-Quarter 1-Dime 1-Nickel 1-Penny
2-Quarters 2-Dimes 4-Pennies
2-$20 1-$10 1-$5 4-$1
1-Nickel
1-Quarter 1-Dime 1-Nickel 1-Penny
3-Quarters
3-Quarters 2-Dimes 4-Pennies
```

Problem #9
60 Points

9. Maite

Program Name: Maite.java

Input File: maite.dat

Test Input File:

Note: This judge data file contains a large number of lines. This page shows the first portion of those lines. See the judge data file for the complete input data listing.

```

35
6 2
1 3 1 3 1 3
7 3
2 7 7 2 7 7 2
9 1
1 4 7 10 3 6 9 2 5
2 1
1 2
2 2
1 2
2 1
1 1
3 1
1 2 1
2 2
-1000000000 1000000000
15 4
-677024483 -448992662 -677024483 -677024483 -677024483 -821385950 490328923 -
677024483 -677024483 490328923 -448992662 490328923 -821385950 -443839343
490328923
17 10
709004888 -818852167 -524804217 -558136903 -818852167 -861583313 177558432
710020886 -558136903 -107242743 710020886 177558432 -818852167 177558432 -
818852167 710020886 -433262369

```

(data continues, see in data file)

Test Output To Screen:

Case #1: 4	Case #19: 6
Case #2: 5	Case #20: 17
Case #3: 9	Case #21: 18
Case #4: 2	Case #22: 19
Case #5: 2	Case #23: 13
Case #6: 1	Case #24: 16
Case #7: 2	Case #25: 15
Case #8: 2	Case #26: 4
Case #9: 12	Case #27: 2
Case #10: 17	Case #28: 4
Case #11: 13	Case #29: 13
Case #12: 16	Case #30: 4
Case #13: 11	Case #31: 100
Case #14: 12	Case #32: 15
Case #15: 19	Case #33: 997
Case #16: 15	Case #34: 149997
Case #17: 7	Case #35: 1
Case #18: 19	

Problem #10
60 Points

10. Mayank

Program Name: Mayank.java

Input File: mayank.dat

Test Input File:

50	1FMSK7DH8LGB41207	2FMPK3K99KBC62036
1LNHM81W72Y623687	1FFEZ1CP2LEA74602	2FMPK3K9XLBA59674
1LNHM81W27Y623687	1FTFW1RG8LFB33556	3C4NJCAB7LT218738
1FTYR14C5WPA36284	1G1FB1RS0L0121994	3C4PDCB87KT870618
1FTVR14C5WPA32648	1GCGSCEN3L1212131	3N1CP5BV4LL500607
111111111111111111	1GCPWCEK8LZ238459	5GAERBKW5LJ236586
1M8GDM9AXKP042788	1GKS2HKK8LR256516	5N1AZ2AJ1LN128053
1M8DGM9AXKP042788	1GTP6EE13L1221277	5NM010001LK012345
1C4PJLLB8LD561344	1GTP6EE19L1150201	JM1NDAB77L0413225
1P4SBHCT8LC186943	1GTU9FET6LZ137988	JM3KFADM4L0738177
1C6SRFFT5LN161882	1GYFZER40LF016956	JM3TCADY2L0412765
1C6SRFKT1LN136838	1GYS4JKJ7LR227071	JN1AZ4EH4LMB23210
1FAP68CF1L5125552	1N6AA1EE3LN504980	JN1BJ1CVXLW258372
1FA6P8CF3L5114995	2C3CCABG1LH103064	KMHRC8A34LU028124
1FA6P8TH4K5204253	2C3CDXHGOKH633880	KMHTG6AF3LU024848
1FMCU0G65LUB23930	2C3CDZBT9LH118923	W04GP6SX8K1077991
1FNSC7DH1LGA80498	2C3CDZL99LH142190	ZACNJAABXLPL34041

Test Output To Screen:

#1: VALID	#18: INVALID 5	#35: VALID
#2: INVALID 8	#19: VALID	#36: VALID
#3: VALID	#20: VALID	#37: INVALID 0
#4: INVALID 4	#21: VALID	#38: VALID
#5: VALID	#22: VALID	#39: VALID
#6: VALID	#23: INVALID 2	#40: VALID
#7: INVALID 7	#24: VALID	#41: INVALID 9
#8: VALID	#25: VALID	#42: VALID
#9: VALID	#26: VALID	#43: VALID
#10: VALID	#27: VALID	#44: VALID
#11: VALID	#28: INVALID 1	#45: INVALID 6
#12: INVALID 3	#29: VALID	#46: VALID
#13: VALID	#30: VALID	#47: VALID
#14: VALID	#31: INVALID X	#48: VALID
#15: VALID	#32: VALID	#49: INVALID X
#16: VALID	#33: VALID	#50: VALID
#17: VALID	#34: VALID	

Problem #11

60 Points

11. Rashi

Program Name: Rashi.java

Input File: rashi.dat

Test Input File:

3,3,4
0,0,0,0
69.33,22.05,0,70.8
9
0,0,0,0
71.95,0,0,0
41.64,0,23.47,28.9
3
67.73,0,0,0
0,0,33.31,0
1.92,0,47.67,0
0,94.71,24.62,0
7,5,6
70.26,20.91,45.92,
52.25,54.55,62.78
31.93,73.24,79.67,
7.65,90.35,13.28
59.07,38.22,64.00,
79.06,76.54,24.79
46.64,8.04,24.30,1
8.86,77.09,98.87
70.94,91.89,80.49,
63.34,72.44,82.05
0,0,0,0,0
0,0,0,0,0
0,0,0,0,0
0,0,0,0,0
0,0,0,0,0
0,0,0,0,0
0,0,0,0,0
0,0,0,0,0
0,69.33,22.05,0,70
.89,0
0,0,0,0,0,0
0,0,0,0,0,0
0,0,0,0,0,0
0,71.95,0,0,0,0
0,41.64,0,23.47,28
.93,0
0,67.73,0,0,0,0
0,0,0,0,0,0
0,0,0,0,0,0
0,0,0,33.31,0,0
0,1.92,0,47.67,0,0
0,0,94.71,24.62,0,
0
0,0,0,0,0,0
0,0,0,0,0,0
0,0,0,0,0,0
0,0,0,0,0,0
0,0,0,0,0,0
0,0,0,0,0,0
70.50,37.96,5.14,7
2.08,30.61,1.69
9.91,25.47,44.90,1
4.91,99.73,19.65
46.76,57.85,2.34,5
6.66,71.04,41.84
22.77,84.80,12.64
67.55,55.61,74.38
21.87,89.54,8.02,8
1.14,3.71,16.14
7,5,8
0,70.26,20.91,45.9
2,52.25,54.55,62.7
8,0
0,31.93,73.24,79.6
7,7.65,90.35,13.28
0
0,59.07,38.22,64.0
0,79.06,76.54,24.7
9,0
0,46.64,8.04,24.30
1,18.86,77.09,98.87
0
0,70.94,91.89,80.4
9,63.34,72.44,82.0
5,0
0,0,0,0,0,0,0,0
60.94,0,0,0,0,0,0,
45,38

[illegible]

2,28.85,87.16,37.5
 22,82.49,51.91,50.3
 5,0
 0,23.13,3.7,7.97,2
 3.39,15.95,62.19,0
 0,0,0,0,0,0,0,0
 28.51,0,0,0,0,0,0,43.18
 54.49,0,69.33,22.0
 5,0,70.89,0,63.96
 15.39,0,0,0,0,0,0,97.82
 0,0,0,0,0,0,0,0
 0,93.71,35.35,67.2
 8,39.95,72.53,5.56
 0,7.78,58.33,26.01
 54.48,66.18,21.29
 ,0
 0,0,0,0,0,0,0,0
 41.58,0,71.95,0,0
 0,0,25.22
 34.20,0,41.64,0,23
 .47,28.93,0,43.68
 5.66,0,67.73,0,0,0
 ,0,96.30
 0,0,0,0,0,0,0,0
 0,39.75,61.22,59.8
 ,19.26,20.44,24.36
 ,0
 0,85.09,75.05,90.1
 7,50.89,24.9,63.59
 ,0
 0,0,0,0,0,0,0,0
 56.98,0,0,0,33.31,
 0,0,5.75
 0.27,0,1.92,0,47.6
 7,0,0,1.67
 2.75,0,0,94.71,24.
 62,0,0,73.95
 0,0,0,0,0,0,0,0
 0,81.17,97.32,94.4
 9,43.63,99.63,77.5
 4,0
 0,28.36,10.9,48.27
 ,48.02,38.71,43.27
 ,0
 0,0,0,0,0,0,0,0
 30.88,0,0,0,0,0,0,0
 88.10
 22.75,0,0,0,0,0,0,0
 75.64
 38.39,0,0,0,0,0,0,0
 37.48
 0,0,0,0,0,0,0,0
 0,62.1,98.18,44.79
 ,36.73,66.47,19.72
 ,0
 0,0,0,0,0,0,0,0
 0,70.50,37.96,5.14
 ,72.08,30.61,1.69,
 0
 0,9.91,25.47,44.90
 ,14.91,99.73,19.65
 ,0
 0,46.76,57.85,2.34
 ,56.66,71.04,41.84
 ,0
 0,22.77,84.80,12.6
 4,67.51,55.61,74.3
 8,0
 0,21.87,89.54,8.02
 ,81.14,3.71,16.14,
 0
 0,0,0,0,0,0,0,0
 5,5,5
 65.67,0,55.14,0,45
 .96
 0,0,0,0,0
 55.17,0,80.11,0,77
 .61

0,0,0,0,0
92.07,0,80.15,0,59
.23
0,0,0,0,0
0,2.30,0,17.14,0
0,0,0,0,0
0,2.97,0,19.03,0
0,0,0,0,0
1.59,0,5.95,0,21.0
3
0,0,0,0,0
28.02,0,99.08,0,11
.63
0,0,0,0,0
89.51,0,51.69,0,64
.54
0,0,0,0,0
0,58.46,0,19.88,0
0,0,0,0,0
0,7.23,0,64.82,0
0,0,0,0,0
47.63,0,9.26,0,4.9
6
0,0,0,0,0
12.35,0,37.93,0,25
.52
0,0,0,0,0
75.49,0,47.92,0,15
.98
2,2,2
12.34,0
0,56.78
0,98.76
54.32,0
3,3,3
1,1,1
1,1,1
1,1,1
2,2,2
2,2,2
2,2,2
3,3,3
3,3,3
3,3,3
10,10,10
90.14,44.53,98.35,
65.68,77.71,0.34,6
8.72,70.64,35.70,0
.29
61.14,55.34,67.06,
89.84,79.41,55.88,
51.38,21.76,97.44,
28.08
11.34,69.75,73.11,
20.28,71.72,15.32,
96.08,64.43,68.08,
82.98
82.83,89.55,21.50,
60.41,29.30,59.52,
77.17,44.98,95.23,
4.40
79.53,86.95,23.33,
21.62,40.12,54.67,
60.42,45.20,59.20,
42.93
19.02,54.05,45.29,
98.87,71.84,93.61,
33.75,32.34,5.10,6
2.98
12.00,97.31,48.16,
34.29,21.82,45.57,
42.36,43.25,70.16,
88.03
95.94,80.81,32.98,
97.87,14.47,21.82,
39.95,6.84,64.45,4
0.27
38.63,72.27,76.47,
19.11,56.36,49.38,
52.34,28.24,65.09,
36.57

6.05, 33.26, 94.06, 1
0.01, 52.78, 75.68, 8
3.88, 84.37, 12.55, 6
8.65
38.56, 86.75, 30.62,
43.41, 56.09, 27.02,
20.07, 60.81, 5.66, 5
8.52
10.81, 23.21, 51.35,
64.82, 72.05, 35.18,
32.91, 15.00, 21.18,
82.57
40.04, 85.01, 15.08,
76.05, 78.31, 40.63,
0.10, 15.87, 71.75, 4
1.51
99.06, 41.24, 44.08,
10.74, 77.06, 49.10,
64.57, 18.50, 3.37, 5
7.43
9.61, 61.82, 64.35, 7
6.03, 25.19, 38.43, 8
9.93, 21.89, 18.16, 6
1.68
60.23, 68.56, 70.50,
35.54, 57.26, 7.03, 8
7.78, 31.98, 60.41, 2
9.40
5.45, 39.62, 0.03, 43
.17, 11.72, 34.27, 5.
87, 6.56, 24.07, 37.8
9
32.96, 28.86, 53.68,
18.51, 42.07, 9.09, 7
4.61, 29.65, 29.72, 1
3.64
78.22, 97.68, 74.80,
26.23, 73.01, 10.86,
81.01, 9.14, 22.20, 5
9.19
26.23, 29.54, 67.88,
2.82, 89.85, 86.25, 6
3.85, 79.82, 98.13, 3
7.08
30.42, 54.91, 23.20,
87.18, 86.84, 34.56,
61.54, 53.23, 20.60,
29.59
59.83, 6.62, 49.67, 6
6.98, 29.62, 41.41, 9
2.49, 45.12, 39.77, 1
7.43
59.87, 23.25, 49.13,
99.36, 62.05, 38.31,
33.25, 71.02, 18.43,
88.03
86.30, 90.00, 39.12,
93.48, 88.29, 75.66,
91.47, 45.37, 39.12,
17.16
51.63, 78.56, 12.01,
17.75, 18.19, 45.95,
2.17, 14.21, 12.76, 6
.62
1.26, 99.47, 9.38, 96
.66, 0.17, 17.99, 30
7.01, 10.10, 41.88, 27.8
1
14.05, 61.28, 98.82,
1.13, 46.43, 10.70, 7
5.67, 31.10, 20.20, 2
1.39
70.85, 91.00, 23.27,
2.94, 43.25, 70.16, 9
4.35, 15.93, 78.45, 1
0.81
48.36, 9.68, 2.69, 9
50.73, 68.74, 74.14, 71.
59, 98.78, 3.29, 46.1
6

UIL – Computer Science Judge’s Packet – State - 2021

27.66,34.81,58.01, 44.19,97.28,30.96, 36.71,77.50,36.68, 24.41 60.52,23.91,17.71, 58.35,67.68,13.36, 60.85,31.57,11.30, 68.39 85.70,58.85,94.59, 33.73,20.77,83.32, 61.62,55.93,42.87, 79.16 27.50,75.85,72.01, 38.90,69.05,63.04, 23.37,35.75,51.39, 9.75 36.99,35.07,68.15, 53.28,42.47,56.56, 57.96,94.70,91.50, 43.91 31.86,74.51,24.57, 5.79,98.77,74.98,3 3.72,51.96,24.56,2 6.35 17.69,89.93,44.05, 25.78,39.38,47.40, 21.75,51.38,29.27, 20.22 16.74,91.82,49.75, 57.37,68.95,59.22, 30.98,75.52,44.30, 55.70 96.69,62.64,73.19, 49.76,68.93,51.80, 0.77,64.93,54.71,4 .96 70.04,45.04,3.22,9 4.08,71.89,53.44,9 7.19,19.39,29.20,7 0.35 5.67,82.22,57.88,6 1.68,83.98,60.85,9 1.50,48.17,42.21,6 2.70 36.56,82.36,67.05, 20.98,54.02,71.82, 71.09,38.53,4.19,9 1.49 38.83,33.85,70.20, 87.42,40.61,48.53, 86.90,18.56,28.54, 27.25 92.96,54.68,3.65,7 1.10,94.48,58.20,2 9.72,41.97,49.40,1 8.71 16.09,45.68,46.29, 21.42,87.18,30.43, 28.82,1.92,62.94,2 1.96 91.32,79.51,73.70, 88.15,70.03,26.57, 43.02,31.60,52.99, 75.52 72.65,48.72,17.15, 46.05,80.71,54.14, 0.59,55.33,20.73,9 7.77	91.70,80.72,19.72, 16.86,39.89,52.76, 50.48,16.57,85.55, 48.17 72.85,17.44,81.20, 0.50,12.45,66.02,3 .26,24.88,98.40,25 .29 62.67,15.48,21.33, 35.13,13.91,55.24, 78.56,9.35,72.70,5 8.72 92.14,9.10,63.26,8 1.48,6.57,67.80,29 .21,80.77,44.65,67 .46 66.87,81.55,61.13, 78.26,93.27,93.65, 61.89,5.11,47.48,2 5.88 34.39,72.68,18.98, 51.80,74.80,45.06, 98.65,33.53,81.63, 72.45 63.77,29.56,60.53, 46.02,0.29,85.54,2 1.50,44.74,67.45,4 0.11 51.86,54.87,63.05, 3.13,15.46,99.08,5 9.44,57.15,60.74,4 0.51 2.87,36.48,49.61,4 1.72,6.52,98.53,17 .21,81.08,38.34,79 .55 80.15,2.16,5.48,95 .84,76.96,78.70,4. 00,73.76,19.30,74. 21 81.19,41.52,73.01, 75.88,18.60,65.46, 47.08,6.12,79.03,9 .51 1.11,7.28,10.30,66 .38,42.16,47.68,62 .58,18.38,28.88,50 .00 42.94,60.62,54.05, 17.72,60.07,83.66, 16.52,99.05,80.31, 45.27 15.70,55.57,6.24,1 0.89,87.25,79.42,6 0.23,15.73,86.75,9 .58 28.67,65.45,74.52, 59.25,1.83,38.75,8 5.08,36.83,86.99,5 9.96 32.14,61.97,44.82, 24.76,53.76,46.18, 49.03,39.18,18.78, 46.64 34.77,28.06,56.65, 90.25,58.52,60.32, 22.63,12.39,94.78, 15.18	57.21,23.70,23.37, 76.63,13.01,2.16,8 3.72,0.83,43.48,9. 71 73.42,37.03,97.82, 71.16,25.04,71.89, 2.49,23.82,91.04,1 9.62 47.02,72.04,64.80, 57.89,89.57,15.51, 53.97,51.54,49.25, 24.90 47.38,57.20,7.61,4 2.47,78.82,81.38,2 5.87,10.28,33.81,2 8.55 62.91,67.74,33.59, 23.90,56.52,57.15, 93.58,9.96,59.62,3 3.21 82.71,11.89,89.94, 53.67,99.81,17.71, 2.82,27.72,28.43,3 4.36 14.41,10.67,15.86, 51.33,21.92,14.72, 56.71,30.64,83.61, 32.00 74.51,78.93,67.13, 71.36,99.07,94.34, 73.68,48.92,54.52, 48.06 93.03,52.26,65.55, 99.59,64.99,92.92, 53.82,98.45,45.32, 64.02 3.35,94.21,68.43,3 0.43,51.76,89.43,7 7.48,70.36,99.56,1 4.30 32.15,6.74,17.57,4 1.01,50.23,72.31,3 6.41,2.62,51.18,89 .79 83.89,54.35,73.91, 27.75,22.40,68.86, 64.27,42.98,57.87, 14.18 75.67,6.78,80.28,4 5.70,80.39,46.06,5 5.29,43.55,37.05,7 5.40 34.27,58.78,76.97, 67.79,39.02,39.58, 45.69,55.62,47.32, 65.10 4.02,4.52,66.80,43 .14,52.17,49.44,72 .94,45.46,75.31,98 .84 15.28,93.39,77.84, 88.57,23.97,37.28, 61.26,86.06,29.43, 84.92 11.91,21.23,70.21, 7.76,0.07,62.09,4. 89,10.76,74.30,94. 63	18.65,74.42,90.05, 16.10,34.96,2.93,7 .90,98.02,81.97,53 .86 70.28,73.57,94.80, 86.57,44.14,27.90, 90.00,82.51,99.95, 87.68 25.14,60.59,35.89, 37.50,2.72,0.17,14 .15,4.72,38.49,99. 70 22.61,4.65,24.33,5 5.85,84.68,23.02,6 7.17,10.86,13.41,1 8.59 43.86,70.15,9.22,2 7.06,73.43,26.09,9 .63,39.08,9.40,22. 49 35.40,27.35,67.29, 37.59,17.87,34.49, 78.70,86.31,7.61,9 9.15 19.32,61.55,96.59, 93.62,65.76,17.55, 26.56,90.07,76.39, 47.23 8.82,68.88,82.68,4 .90,78.13,81.40,61 .40,14.93,56.72,82 .11 26.55,0.56,75.94,5 6.72,8.39,80.22,75 .29,13.71,17.43,79 .87 71.69,14.77,61.49, 55.46,34.11,31.00, 20.71,1.00,21.57,4 7.82 55.94,67.17,21.72, 17.84,56.42,69.21, 30.24,0.22,55.74,3 5.60 99.11,13.87,42.00, 10.21,50.17,72.82, 83.06,95.14,35.92, 10.04 5.75,64.20,0.36,98 .64,85.47,37.64,53 .50,38.58,40.36,26 .32 64.00,72.32,72.54, 47.03,56.90,16.16, 91.37,51.01,21.18, 14.56 0.20,45.48,30.82,9 3.92,70.57,9.75,64 .26,7.16,24.48,66. 85 55.31,16.61,55.14, 90.63,43.72,9.44,2 0.32,62.29,81.47,1 6.23 81.08,30.86,51.81, 15.37,33.40,78.94, 10.60,10.80,87.52, 46.24	15.13,61.06,5.86,5 9.61,18.97,49.23,9 5.48,93.85,0.64,81 .16 35.39,54.83,89.92, 4.08,8.15,48.14,73 .87,62.65,80.69,57 .09 12.64,70.66,84.30, 82.88,14.14,56.49, 15.94,20.10,6.40,1 5.90 5,5,5 90.14,0,98.35,0,77 .71 0,0,0,0,0 0,0,0,0,0 0,0,0,0,0 0,0,0,0,40.12 19.02,0,45.29,0,71 .84 12.00,0,0,0,21.82 0,0,0,0,0 0,0,0,0,0 0,0,0,10.01,52.78 0,0,30.62,0,0 10.81,0,51.35,0,72 .05 40.04,0,0,0,78.31 0,0,0,0,0 0,0,64.35,76.03,0 0,0,0,0,0 0,39.62,0.03,0,0 32.96,0,0,18.51,42 .07 78.22,0,0,0,0 0,29.54,67.88,0,0 0,0,0,0,0 0,6.62,0,0,0 0,23.25,49.13,99.3 6,0 86.30,0,0,0,0 51.63,78.56,0,0,0 5,5,5 0,0,0,0,0 0,0,0,0,0 0,0,0,0,0 0,0,0,0,0 0,0,0,0,0 0,0,0,0,0 0,0,14.93,0,0 0,0,0,0,0 0,0,95.14,0,0 0,0,38.58,0,0 0,0,51.01,0,0 0,0,0,0,0 0,0,0,0,0 0,0,0,0,0 0,0,0,0,0 0,0,0,0,0
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Test Output To Screen:

TEST CASE #1:
323.60
274.62
=====

TEST CASE #2:
1679.42
1247.17
323.60
274.62
=====

TEST CASE #3:
1679.42

1247.17
864.14
524.02
323.60
274.62
=====

TEST CASE #4:
1699.26
1679.42
1349.14
1247.17
864.14

524.02
323.60
274.62
=====

TEST CASE #5:
NO CLUSTERS FOUND
=====

TEST CASE #6:
NO CLUSTERS FOUND
=====

TEST CASE #7:
54.00

=====

TEST CASE #8:
48379.50
=====

TEST CASE #9:
840.39
825.93
=====

TEST CASE #10:
257.02
=====

Problem #12
60 Points

12. Sergio

Program Name: Sergio.java

Input File: sergio.dat

Test Input File:

a14	5	18 19	13 7
1	4	19 0	14 6
0	0 1	20	15 5
1	0 2	19	16 4
1	0 3	0 1	17 3
0 0	0 4	1 2	18 2
2	5	2 3	19 1
2	10	3 4	20 0
0 1	0 1	4 5	21
1 0	1 2	5 6	25
2	3 2	6 7	12
1	3 4	7 8	6
0 1	4 2	8 9	8
3	0 2	9 10	18
3	1 4	10 11	20
0 0	0 3	11 12	7
1 1	0 4	12 13	7
2 2	1 3	13 14	13
3	20	14 15	2
2	20	15 16	1
0 1	0 1	16 17	10
1 2	1 2	17 18	16
3	2 3	18 19	15
9	3 4	21	6
0 0	4 5	21	19
0 1	5 6	0 20	12
0 2	6 7	1 19	12
1 0	7 8	2 18	9
1 1	8 9	3 17	8
1 2	9 10	4 16	20
2 0	10 11	5 15	2
2 1	11 12	6 14	12
2 2	12 13	7 13	5
3	13 14	8 12	3
3	14 15	9 11	14
1 0	15 16	10 10	12
2 0	16 17	11 9	
2 1	17 18	12 8	

Test Output To Screen:

No cycle
Cycle
Cycle
No cycle
Cycle
No cycle
Cycle
No cycle
No cycle
No cycle
Cycle
No cycle
Cycle
Cycle